

DEPARTMENT OF MATHEMATICS, IIT - GUWAHATI

Deferred Semester of the Academic Year 2023-2024

MA 101 Mathematics I

Problem Sheet 5: Line integrals, Multiple integrals and applications, Green's Theorem, Stokes Theorem and Divergence Theorem.

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1. Calculate the line integral of the vector field along the path described:

- (a) $f(x, y) = (x^2 + y^2)\mathbf{i} + (x^2 - y^2)\mathbf{j}$ from $(0, 0)$ to $(2, 0)$ along the curve $y = 1 - |1 - x|$
- (b) $f(x, y, z) = 2xy\mathbf{i} + (x^2 + z)\mathbf{j} + (y + z)\mathbf{k}$ from $(1, 0, 2)$ to $(3, 4, 1)$ along a line segment
- (c) $f(x, y, z) = x\mathbf{i} + y\mathbf{j} + (xz - y)\mathbf{k}$, along the path described by $\alpha(t) = t^2\mathbf{i} + 2t\mathbf{j} + 4t^3\mathbf{k}$, $0 \leq t \leq 1$.

2. Find the line integral of $f(x, y, z) = z$ with respect to arc length of the curve given by

$$\mathbf{r}(t) = (t \cos t)\mathbf{i} + (t \sin t)\mathbf{j} + (t)\mathbf{k}, \quad 0 \leq t \leq 1.$$

3. For each of the following vector fields show that f is not a gradient vector in $\mathbf{R}^2$. Then for each of the following find a closed path C such that $\oint_C f \neq 0$ and if possible find a closed path C such that $\oint_C f = 0$.

- (a) $f(x, y) = y\mathbf{i} - x\mathbf{j}$
- (b) $f(x, y) = \frac{y}{(x^2 + y^2)}\mathbf{i} - \frac{x}{(x^2 + y^2)}\mathbf{j}$, for $(x, y) \neq (0, 0)$.

4. Show that each of the following functions F is a gradient vector and find an f for each F such that $F = \nabla f$.

- (a) $F(x, y) = 3x^2y\mathbf{i} + x^3\mathbf{j}$
- (b) $F(x, y) = (\sin y - y \sin x + x)\mathbf{i} + (\cos x + x \cos y + y)\mathbf{j}$.

5. Use Green's theorem to evaluate the line integral along the given positively oriented curve:

- (a) $\int_C (y + e^{\sqrt{x}})dx + (2x + \cos y^2)dy$, C is the boundary of the region enclosed by the parabolas $y = x^2$ and $x = y^2$
- (b) $\int_C xydx + 2x^2dy$, C consists of the line segment from $(-2, 0)$ to $(2, 0)$ and top half of the circle $x^2 + y^2 = 4$.

6. Use Green's theorem to find out the work done by the force $\mathbf{F}(x, y) = x(x + y)\mathbf{i} + xy^2\mathbf{j}$ in moving a particle from the origin along x -axis to $(1, 0)$ and then along the line segment to $(0, 1)$, and then back to the origin along y -axis.

7. Let D be a region bounded by a simple closed path C in the xy -plane. Use Green's theorem to prove that the coordinates of the centroid $(\bar{x}, \bar{y})$ of D are

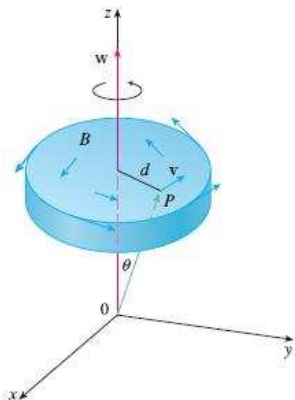
$$\bar{x} = \frac{1}{2A} \oint x^2 dy, \quad \bar{y} = -\frac{1}{2A} \oint y^2 dx \quad \text{where } A \text{ is the area of } D.$$

8. If $\mathbf{r}(x, y) = x\mathbf{i} + y\mathbf{j}$ and $r = |\mathbf{r}|$, let

$$f(x, y) = \frac{\partial(\log r)}{\partial y}\mathbf{i} - \frac{\partial(\log r)}{\partial x}\mathbf{j}$$

for $r > 0$.

Let C be a smooth simple closed curve in the annulus $1 < x^2 + y^2 < 25$, then find all possible values of the line integral of f along C .



9. The exercise demonstrates a connection between curl vector and rotations. Let $\mathbf{B}$ be a rigid body rotating about z -axis. The rotation can be described by the vector $\mathbf{w} = \omega \mathbf{k}$, where ω is the angular speed of $\mathbf{B}$, that is, the tangential speed at any point P in B divided by the distance d from the axis of rotation. Let $\mathbf{r} = \langle x, y, z \rangle$ be the position vector of P .

- By considering the angle θ in the figure, show that the velocity field of B is given by $\mathbf{v} = \mathbf{w} \times \mathbf{r}$
- Show that $\mathbf{v} = -\omega y \mathbf{i} + \omega x \mathbf{j}$
- Show that $\nabla \times \mathbf{v} = 2\mathbf{w}$.

10. Use Stokes' Theorem to evaluate

- $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ where $\mathbf{F} = xyz\mathbf{i} + xy\mathbf{j} + x^2yz\mathbf{k}$ and S consists of the top and the four sides (but not the bottom) of the cube with vertices $(\pm 1, \pm 1, \pm 1)$, oriented outward
- $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = 2z\mathbf{i} + 4x\mathbf{j} + 5y\mathbf{k}$ and C is the curve of intersection of the plane $z = x + 4$ and the cylinder $x^2 + y^2 = 4$.

11. Calculate the work done by the force field

$$\mathbf{F}(x, y, z) = (x^x + z^2)\mathbf{i} + (y^y + x^2)\mathbf{j} + (z^z + y^2)\mathbf{k}$$

when a particle moves under its influence around the edge of the part of the sphere $x^2 + y^2 + z^2 = 4$ that lies in the first octant, in a counterclockwise direction as viewed from above.

12. Let S be the surface of the solid cylinder T bounded by the planes $z = 0$ and $z = 3$ and the cylinder $x^2 + y^2 = 4$. Calculate the outward flux $\iint_S \mathbf{F} \cdot \mathbf{n} dS$ given $\mathbf{F}(x, y, z) = (x^2 + y^2 + z^2)(x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$.

13. Use Divergence Theorem to evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} dS$ where

$$\mathbf{F}(x, y, z) = z^2 x \mathbf{i} + \left(\frac{1}{3} y^3 + \tan z \right) \mathbf{j} + (x^2 z + y^2) \mathbf{k}$$

and S is the top half of the sphere $x^2 + y^2 + z^2 = 1$.